

Spring Data JPA - Quick Example

Table Creation

```
CREATE DATABASE ormlearn;
```

```
USE ormlearn;
```

```
CREATE TABLE country(  
    code VARCHAR(2) PRIMARY KEY,  
    name VARCHAR(50)  
);
```

```
INSERT INTO country VALUES('IN','India');  
INSERT INTO country VALUES('US','United States');
```

application.properties

```
spring.datasource.url=jdbc:mysql://localhost:3306/ormlearn
```

```
spring.datasource.username=root
```

```
spring.datasource.password=root
```

```
spring.jpa.hibernate.ddl-auto=validate
```

```
spring.jpa.show-sql=true
```

```
logging.level.org.hibernate.SQL=debug
```

Country.java

```
package com.cognizant.ormlearn.model;
```

```
import jakarta.persistence.*;
```

```
@Entity
```

```
@Table(name="country")
```

```
public class Country {

    @Id
    @Column(name="code")
    private String code;

    @Column(name="name")
    private String name;

    public Country() {
    }

    public Country(String code, String name) {
        this.code = code;
        this.name = name;
    }

    public String getCode() {
        return code;
    }

    public void setCode(String code) {
        this.code = code;
    }

    public String getName() {
        return name;
    }
}
```

```

    public void setName(String name) {
        this.name = name;
    }

    @Override
    public String toString() {
        return "Country [code=" + code + ", name=" + name + "]";
    }
}

```

CountryRepository.java

```

package com.cognizant.ormlearn.repository;

import org.springframework.data.jpa.repository.JpaRepository;
import org.springframework.stereotype.Repository;

import com.cognizant.ormlearn.model.Country;

@Repository
public interface CountryRepository extends JpaRepository<Country, String> {

}

```

CountryService.java

```

package com.cognizant.ormlearn.service;

```

```
import java.util.List;
```

```
import org.springframework.beans.factory.annotation.Autowired;
```

```
import org.springframework.stereotype.Service;
```

```
import com.cognizant.ormlearn.model.Country;
```

```
import com.cognizant.ormlearn.repository.CountryRepository;
```

```
import jakarta.transaction.Transactional;
```

```
@Service
```

```
public class CountryService {
```

```
    @Autowired
```

```
    private CountryRepository repository;
```

```
    @Transactional
```

```
    public List<Country> getAllCountries() {
```

```
        return repository.findAll();
```

```
    }
```

```
    @Transactional
```

```
    public Country getCountry(String code) {
```

```
        return repository.findById(code).orElse(null);
```

```
    }
```

```
    @Transactional
```

```
public void addCountry(Country country) {  
    repository.save(country);  
}
```

```
@Transactional
```

```
public void updateCountry(String code, String name) {
```

```
    Country country = repository.findById(code).orElse(null);
```

```
    if(country != null){  
        country.setName(name);  
        repository.save(country);  
    }
```

```
}
```

```
@Transactional
```

```
public void deleteCountry(String code){
```

```
    repository.deleteByld(code);
```

```
}
```

```
}
```

OrmLearnApplication.java

```
package com.cognizant.ormlearn;
```

```
import org.springframework.boot.SpringApplication;
```

```
import org.springframework.boot.autoconfigure.SpringBootApplication;
```

```
import org.springframework.context.ApplicationContext;
```

```
import com.cognizant.ormlearn.model.Country;
```

```
import com.cognizant.ormlearn.service.CountryService;

@SpringBootApplication
public class OrmLearnApplication {

    public static void main(String[] args) {

        ApplicationContext context =
            SpringApplication.run(OrmLearnApplication.class, args);

        CountryService service =
            context.getBean(CountryService.class);

        System.out.println("All Countries");
        service.getAllCountries().forEach(System.out::println);

        System.out.println("\nFind Country");
        System.out.println(service.getCountry("IN"));

        System.out.println("\nAdd Country");
        service.addCountry(new Country("JP","Japan"));

        System.out.println(service.getCountry("JP"));

        System.out.println("\nUpdate Country");
        service.updateCountry("JP","Japan Updated");

        System.out.println(service.getCountry("JP"));
```

```

        System.out.println("\nDelete Country");

        service.deleteCountry("JP");

    }

}

```

```

<terminated> OrmLearnApplication [Java Application] C:\Program Files\Java\jdk-17\bin\javaw.exe (30-Jun-2026, 8:36:51 am - 8:36:56 am)

:: Spring Boot :: (v4.1.0)

2026-06-30T08:36:53.434+05:30 INFO 21596 --- [ restartedMain] c.c.orm_learn.OrmLearnApplication : Starting OrmLearnApplication using Java 17.0.7 with PID 21596 (D:\orm-learn\orm-learn\target\
2026-06-30T08:36:53.453+05:30 INFO 21596 --- [ restartedMain] c.c.orm_learn.OrmLearnApplication : No active profile set, falling back to 1 default profile: "default"
2026-06-30T08:36:53.505+05:30 INFO 21596 --- [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : Devtools property defaults active! Set 'spring.devtools.add-properties' to 'false' to disable
2026-06-30T08:36:53.909+05:30 INFO 21596 --- [ restartedMain] .s.d.r.c.RepositoryConfigurationDelegate : Bootstrapping Spring Data JPA repositories in DEFAULT mode.
2026-06-30T08:36:53.961+05:30 INFO 21596 --- [ restartedMain] .s.d.r.c.RepositoryConfigurationDelegate : Finished Spring Data repository scanning in 43 ms. Found 1 JPA repository interface.
2026-06-30T08:36:54.347+05:30 INFO 21596 --- [ restartedMain] org.hibernate.orm.jpa : HH000540: Processing PersistenceUnitInfo [name: default]
2026-06-30T08:36:54.414+05:30 INFO 21596 --- [ restartedMain] org.hibernate.orm.core : HH000001: Hibernate ORM core version 7.4.1.Final
2026-06-30T08:36:54.867+05:30 INFO 21596 --- [ restartedMain] o.s.o-j-p.SpringPersistenceUnitInfo : No LoadTimeWeaver setup: ignoring JPA class transformer
2026-06-30T08:36:54.899+05:30 INFO 21596 --- [ restartedMain] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Starting...
2026-06-30T08:36:55.224+05:30 INFO 21596 --- [ restartedMain] com.zaxxer.hikari.pool.HikariPool : HikariPool-1 - Added connection com.mysql.cj.jdbc.ConnectionImpl@1a05b29f
2026-06-30T08:36:55.228+05:30 INFO 21596 --- [ restartedMain] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Start completed.
2026-06-30T08:36:55.341+05:30 INFO 21596 --- [ restartedMain] org.hibernate.orm.connections.pooling : HH000005: Database info:
Database JDBC URL [jdbc:mysql://localhost:3306/ormlearn]
Database driver: MySQL Connector/J
Database dialect: MySQLDialect
Database version: 8.0.27
Default catalog/schema: ormlearn/undefined
Autocommit mode: undefined/unknown
Isolation level: REPEATABLE_READ [default REPEATABLE_READ]
JDBC fetch size: none
Pool: DataSourceConnectionProvider
Minimum pool size: undefined/unknown
Maximum pool size: undefined/unknown
2026-06-30T08:36:56.114+05:30 INFO 21596 --- [ restartedMain] org.hibernate.orm.core : HH000489: No JTA platform available (set 'hibernate.transaction.jta.platform' to enable JTA |
2026-06-30T08:36:56.166+05:30 INFO 21596 --- [ restartedMain] j.LocalContainerEntityManagerFactoryBean : Initialized JPA EntityManagerFactory for persistence unit 'default'
2026-06-30T08:36:56.257+05:30 INFO 21596 --- [ restartedMain] o.s.d.j-r.query.QueryEnhancerFactories : Hibernate is in classpath; If applicable, HQL parser will be used.
2026-06-30T08:36:56.525+05:30 INFO 21596 --- [ restartedMain] c.c.orm_learn.OrmLearnApplication : Started OrmLearnApplication in 3.469 seconds (process running for 4.06)
All Countries
2026-06-30T08:36:56.691+05:30 DEBUG 21596 --- [ restartedMain] org.hibernate.SQL : select c1_0.code,c1_0.name from country c1_0
Hibernate: select c1_0.code,c1_0.name from country c1_0
Country [code=IN, name=India]
Country [code=US, name=United States]

Find Country
2026-06-30T08:36:56.730+05:30 DEBUG 21596 --- [ restartedMain] org.hibernate.SQL : select c1_0.code,c1_0.name from country c1_0 where c1_0.code=?
Hibernate: select c1_0.code,c1_0.name from country c1_0 where c1_0.code=?
Country [code=IN, name=India]

Add Country
2026-06-30T08:36:56.737+05:30 DEBUG 21596 --- [ restartedMain] org.hibernate.SQL : select c1_0.code,c1_0.name from country c1_0 where c1_0.code=?
Hibernate: select c1_0.code,c1_0.name from country c1_0 where c1_0.code=?
2026-06-30T08:36:56.748+05:30 DEBUG 21596 --- [ restartedMain] org.hibernate.SQL : insert into country (name,code) values (?,?)
2026-06-30T08:36:56.782+05:30 DEBUG 21596 --- [ restartedMain] org.hibernate.SQL : select c1_0.code,c1_0.name from country c1_0 where c1_0.code=?
Hibernate: select c1_0.code,c1_0.name from country c1_0 where c1_0.code=?
Country [code=JP, name=Japan]

Update Country
2026-06-30T08:36:56.785+05:30 DEBUG 21596 --- [ restartedMain] org.hibernate.SQL : select c1_0.code,c1_0.name from country c1_0 where c1_0.code=?
Hibernate: select c1_0.code,c1_0.name from country c1_0 where c1_0.code=?
2026-06-30T08:36:56.789+05:30 DEBUG 21596 --- [ restartedMain] org.hibernate.SQL : update country set name=? where code=?
2026-06-30T08:36:56.797+05:30 DEBUG 21596 --- [ restartedMain] org.hibernate.SQL : select c1_0.code,c1_0.name from country c1_0 where c1_0.code=?
Hibernate: select c1_0.code,c1_0.name from country c1_0 where c1_0.code=?
Country [code=JP, name=Japan Updated]

Delete Country
2026-06-30T08:36:56.800+05:30 DEBUG 21596 --- [ restartedMain] org.hibernate.SQL : select c1_0.code,c1_0.name from country c1_0 where c1_0.code=?
Hibernate: select c1_0.code,c1_0.name from country c1_0 where c1_0.code=?
2026-06-30T08:36:56.804+05:30 DEBUG 21596 --- [ restartedMain] org.hibernate.SQL : delete from country where code=?
2026-06-30T08:36:56.816+05:30 INFO 21596 --- [ionShutdownHook] j.LocalContainerEntityManagerFactoryBean : Closing JPA EntityManagerFactory for persistence unit 'default'
2026-06-30T08:36:56.819+05:30 INFO 21596 --- [ionShutdownHook] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Shutdown initiated...
2026-06-30T08:36:56.826+05:30 INFO 21596 --- [ionShutdownHook] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Shutdown completed.

```

Hands on 4 Difference between JPA, Hibernate and Spring Data JPA

application.properties

```
spring.datasource.url=jdbc:mysql://localhost:3306/employee_db
```

```
spring.datasource.username=root  
spring.datasource.password=yourpassword
```

```
spring.jpa.hibernate.ddl-auto=update  
spring.jpa.show-sql=true  
spring.jpa.properties.hibernate.format_sql=true  
server.port=8081
```

Database Creation

```
CREATE DATABASE employee_db;
```

Employee.java

```
package com.library.entity;
```

```
import jakarta.persistence.*;
```

```
@Entity
```

```
@Table(name="employees")
```

```
public class Employee {
```

```
    @Id
```

```
    @GeneratedValue(strategy = GenerationType.IDENTITY)
```

```
    private Integer id;
```

```
    private String name;
```

```
    private double salary;
```



```
public Employee() {  
}
```

```
public Employee(String name, double salary) {  
    this.name = name;  
    this.salary = salary;  
}
```

```
public Integer getId() {  
    return id;  
}
```

```
public void setId(Integer id) {  
    this.id = id;  
}
```

```
public String getName() {  
    return name;  
}
```

```
public void setName(String name) {  
    this.name = name;  
}
```

```
public double getSalary() {  
    return salary;  
}
```

```
    public void setSalary(double salary) {  
        this.salary = salary;  
    }  
}
```

EmployeeRepository.java

```
package com.library.repository;
```

```
import org.springframework.data.jpa.repository.JpaRepository;
```

```
import com.library.entity.Employee;
```

```
public interface EmployeeRepository extends JpaRepository<Employee, Integer> {  
  
}
```

EmployeeService.java

```
package com.library.service;
```

```
import org.springframework.beans.factory.annotation.Autowired;
```

```
import org.springframework.stereotype.Service;
```

```
import org.springframework.transaction.annotation.Transactional;
```

```
import com.library.entity.Employee;
```

```
import com.library.repository.EmployeeRepository;
```

```
@Service
```

```
public class EmployeeService {
```

@Autowired

private EmployeeRepository employeeRepository;

@Transactional

public void addEmployee(Employee employee) {

 employeeRepository.save(employee);

}

}

EmployeeController.java

package com.library.controller;

import org.springframework.beans.factory.annotation.Autowired;

import org.springframework.web.bind.annotation.*;

import com.library.entity.Employee;

import com.library.service.EmployeeService;

@RestController

public class EmployeeController {

@Autowired

private EmployeeService employeeService;

@PostMapping("/employee")

public String addEmployee(@RequestBody Employee employee) {

 employeeService.addEmployee(employee);

 return "Employee Saved Successfully";

```
}
}
```

